Sabal Katuwal

Computer Engineer
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EDUCATION

Kathmandu University, Nepal 2019-2024
Bachelor of Engineering in Computer Engineering
CGPA : 3.66/4

Hetauda School of Management, Nepal 2017-2019
Higher Secondary Education in Science (Grade 11 and 12)
CGPA : 3.29/4

Solidarity International Academy, Nepal 2017
Secondary Education Examination (SEE)
GPA : 3.75/4

EXPERIENCE

Software Engineer | Nepal Clearing House Ltd. | Nepal | June 2025 - Present *(Full time, 40 hours/week)*

- Developed a secure digital mobile banking application using SwiftUI, UIKit, and modern iOS frameworks, with a strong focus on mobile security, data protection, and encryption standards.
- Improved the security and stability of the ConnectIPS app by resolving critical vulnerabilities and performance issues, resulting in a 30% increase in overall reliability.
- Used secure coding practices with Keychain, RSA, and jailbreak checks to ensure financial-grade app security and safe API communication.

Associate Software Engineer | Gurzu Inc. | Nepal | Jan 2025 - June 2025 *(Full time, 40 hours/week)*

- Implemented NFC and BLE communication for secure, low-latency interaction with wearable devices on iOS.
- Ensured efficient memory management and robust handling of device communication, maintaining a stable and responsive user experience.
- Develop and optimize features, improving app performance and user experience.

Researcher | Nepal Rastra Bank | Nepal | April 2024 - Jan 2025 *(Full time, 40 hours/week)*

- Contributed in high-performance, secure mobile applications for the retail Central Bank Digital Currency (CBDC) ecosystem.
- Conducted requirement analysis, system design, and technical research to build a cyber-resilient CBDC infrastructure with strong cryptographic and operational security controls.
- Analyzed implementation challenges, such as scalability, offline payment feasibility, resilience, and regulatory compliance and proposed practical solutions tailored to Nepal's financial ecosystem.

SELECTED PROJECTS

Central Bank Digital Currency (CBDC) System Design & Prototype

- Conducted in-depth research on CBDC design principles, including ledger architecture, consensus mechanisms, UTXO vs. account-based models, privacy considerations, and interoperability requirements.
- Contributed to the development of the initial prototype CBDC system, applying public-key cryptography, digital signatures, X.509 certificates, and secure transaction flows to ensure authenticity, integrity, and auditability.
- Designed and implemented a secure mobile wallet for digital currency transactions, facilitating the adoption of digital currency and improving transaction efficiency.

One OS (Appstore) ***(Wearable Payment System, Mobile Application)***

- Built a secure app enabling NFC-based pairing with wearable devices and virtual payment card linking, ensuring encrypted and seamless contactless transactions.
- Designed a scalable MVVM-C architecture with strong communication security, integrating banking APIs for reliable, fast, and protected fund transfers.

Connect Ips (AppStore) ***(Interbank Clearing & Payment Platform, iOS App)***

- Implemented jailbreak detection and strengthened credential security using Keychain and Secure Enclave.
- Reinforced secure transaction flow with robust certificate pinning and strict API request validation.

Digital Mobile Banking System ***(Nepal Unified Mobile Banking App)***

- Implemented fund transfer, offline payment, and cross-border transaction features, improving reliability and user experience.

- Integrated VPN detection at the network layer and restricted sensitive features, resulting in stronger app security and reduced exposure to unsafe network environments.

Bright ACCA (Appstore) ***(Secure Student Learning iOS Platform)***

- Built using SwiftUI and MVVM architecture, ensuring a scalable and maintainable codebase with smooth performance.
- Implemented secure authentication using Keychain, biometric login, and encrypted local storage (SwiftData).
- Added key features such as an MCQ engine, discussion section, notes system, and real-time updates through push notifications and socket-based communication.

TECHNICAL SKILLS

Cryptography Technology: RSA, AES, TLS/SSL, key exchange, certificate pinning, Secure Enclave, MITM Prevention
Mobile Security Expertise: secure coding for financial apps, CBDC security research, secure NFC/BLE protocols, API hardening, jailbreak detection, secure storage
Programming Languages: Swift, Dart, Python, C, C++, SQL, JavaScript
Tools: Xcode, Postman, Charles Proxy, Wireshark, Burp Suite (basic), Core ML, NumPy
Databases: MySQL, Firebase, SQLite, Dbeaver
Cloud / DevOps: AWS, DevOps, GitLab CI, GitHub Actions
Version Control: Git, GitHub, Gitlab
Operating Systems: Linux, Windows, MacOS
Document Creation: Microsoft Office Suite (MS Excel, MS Word, MS Powerpoint)

TECHNICAL CERTIFICATIONS AND PRESENTATIONS

- AWS Training (Genese Academy).
- Conducted literature review and delivered technical paper presentations on “Cyber Resilience of the Central Bank Digital Currency Ecosystem.” (Presentation Link)
- Solo presentation and article review on “Why Ransomware Needs A Human Touch” written by Jürgen Bohn et al., as a mini project of Computer Ethics course, sixth semester. Link to the article: https://ieeexplore.ieee.org/abstract/document/8585650. (My Review Report Link)
- Joint group presentation: “Load Balancing in Cloud” as a part of Cloud Computing elective course, seventh semester.
- Joint group presentation on our startup idea: “Business Plan - SahaYatri, an electronic payment system for bus fares” as a project of Entrepreneurship Development course, eighth semester.

SELECTED LEADERSHIP AND VOLUNTEERING SERVICES

Founding President - Kathmandu University Indoor Games Club (2023)

- Founded and led a 30+ member organization; organized national-level indoor sports events and professional training workshops.

Mister Nepal 2025 – Most Creative Title, Top 7 Finalist (2025)

- Selected among 4,500 participants nationwide; completed intensive leadership, communication, and personality development training throughout the competition. (Certificate)

Event Organizer - Tecno Fair, Nepal Technology Innovation Center (2020)

- Managed event logistics, volunteer coordination, and industry-academia collaboration for startup showcases, technology exhibitions, and innovation programs.

Executive Member - Kathmandu University Robotics Club (2020)

- Supported technical workshops, project coordination, and club operations, contributing to robotics initiatives and student-led engineering activities..

EXTRACURRICULARS

Master of Ceremony - IT Meet 2023

- Hosted major university tech events, coordinating with organizers, professionals, and industry guests.

Sports Achievements

- Winner, KUCC Football Tournament (2023)
- First Runner-Up, Badminton Doubles - KU National-Level Grand Indoor Tournament
- First Runner-Up, Badminton Singles - KUCC Tournament (2022, 2023)

Competitions & Hackathons

- First Runner-Up - Idea Contest 2023 (organised by KOICA & NTIC)
- Participant, KU HackFest (2023), contributing to rapid prototyping and collaborative problem-solving.

Advisory Member - Kathmandu University Indoor Games Club

- Provided strategic guidance for club activities, competitions, and event planning